Wave Reflection and Transmission at Oblique Incidence - Perpendicular Polarization - Part 2

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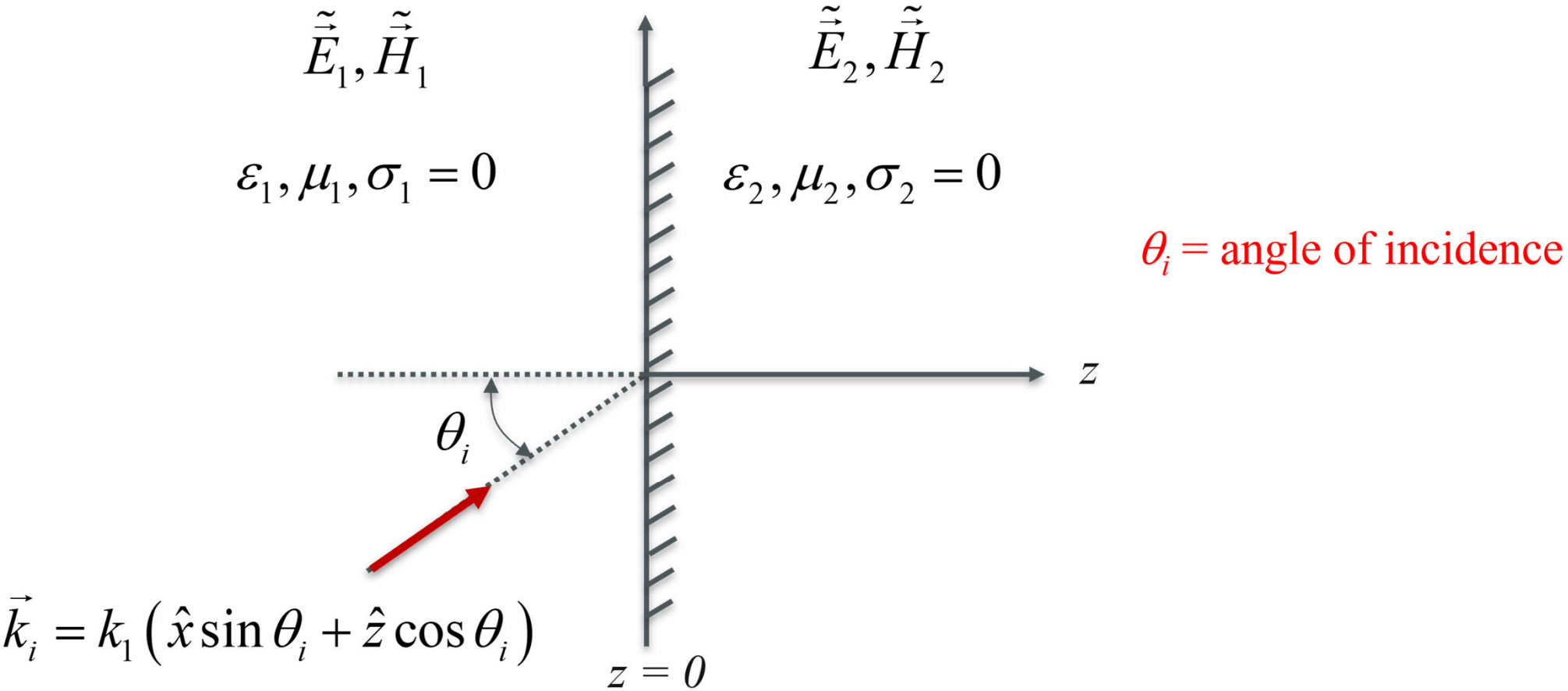
Lecture Objective

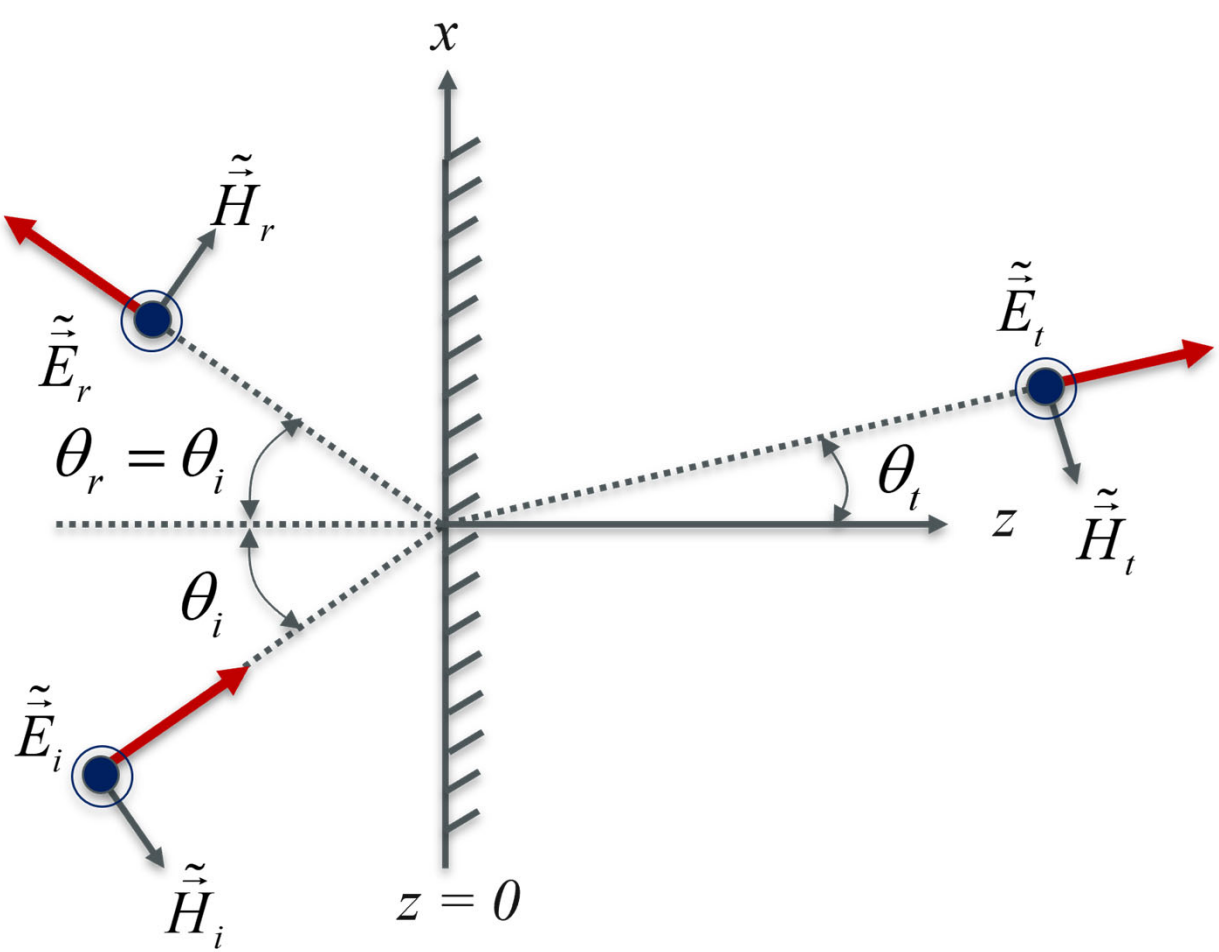
By the end of this lecture, you will be prepared to:

- Analyze plane wave reflection at the boundary between two dielectric half-spaces at oblique incidence for a wave with perpendicular polarization.

Oblique Incidence On a Plane Dielectric Boundary



Incident, Reflected and Transmitted Waves for Perpendicular Polarization



θ_i = angle of incidence

θ_r = angle of reflection

θ_t = angle of transmission

$$\vec{k}_i = k_1\left(\hat{x}\sin\theta_i + \hat{z}\cos\theta_i\right)$$

$$\vec{k}_r = k_1\left(\hat{x}\sin\theta_i - \hat{z}\cos\theta_i\right)$$

$$\vec{k}_t = k_2\left(\hat{x}\sin\theta_t + \hat{z}\cos\theta_t\right)$$

Incident Wave for Perpendicular Polarization

- The electric field of the incident wave is given by

$$\tilde{\vec{E}}_i = \hat{y}\, E_0^i\, e^{-j\vec{k}_i \cdot \vec{R}} = \hat{y}\, E_0^i\, e^{-jk_1\left(x\sin\theta_i + z\cos\theta_i\right)}$$

- The magnetic field of the incident wave is given by

$$\tilde{\vec{H}}_i = \frac{1}{\eta_1}\hat{k}_i \times \tilde{\vec{E}}_i = \frac{1}{\eta_1}(\hat{x}\sin\theta_i + \hat{z}\cos\theta_i) \times \tilde{\vec{E}}_i$$

$$= (-\hat{x}\cos\theta_i + \hat{z}\sin\theta_i)\frac{E_0^i}{\eta_1} e^{-jk_1(x\sin\theta_i + z\cos\theta_i)}$$

Reflected Wave for Perpendicular Polarization

- Postulate that the electric field of the reflected wave is given by

$$\tilde{\vec{E}}_r = \hat{y}\, E_0^r\, e^{-j\vec{k}_r \bullet \vec{R}} = \hat{y}\, E_0^r\, e^{-jk_1\left(x\sin\theta_i - z\cos\theta_i\right)}$$

- The magnetic field of the reflected wave is given by

$$\tilde{\vec{H}}_r = \frac{1}{\eta_1}\hat{k}_r \times \tilde{\vec{E}}_r = \frac{1}{\eta_1}(\hat{x}\sin\theta_i - \hat{z}\cos\theta_i) \times \tilde{\vec{E}}_r$$

$$= (\hat{x}\cos\theta_i + \hat{z}\sin\theta_i)\frac{E_0^r}{\eta_1} e^{-jk_1(x\sin\theta_i - z\cos\theta_i)}$$

Transmitted Wave for Perpendicular Polarization

- Postulate that the electric field of the transmitted wave is given by

$$\overline{E}_t = \hat{y}\, E_0^t\, e^{-j\vec{k}_t \bullet \vec{R}} = \hat{y}\, E_0^t\, e^{-jk_2\left(x\sin\theta_t + z\cos\theta_t\right)}$$

- The magnetic field of the transmitted wave is given by

$$\tilde{\vec{H}}_t = \frac{1}{\eta_1}\hat{k}_t \times \tilde{\vec{E}}_t = \frac{1}{\eta_1}(\hat{x}\sin\theta_t + \hat{z}\cos\theta_t) \times \tilde{\vec{E}}_t$$

$$= (-\hat{x}\cos\theta_t + \hat{z}\sin\theta_t)\frac{E_0^t}{\eta_2}e^{-jk_2(x\sin\theta_t + z\cos\theta_t)}$$

Reflected and Transmitted Waves for Oblique Incidence On a Plane Dielectric Boundary

- To uniquely specify the reflected and transmitted waves in terms of the incident wave, we need to evaluate:
  - E_0^r
  - E_0^t
- These unknowns are obtained by enforcing continuity of the tangential electric and magnetic fields across the boundary at $z = 0$ for all values of x.

Reflected and Transmitted Waves for Perpendicular Polarization

- Continuity of tangential electric field:

$$E_{1y}(x,0) = E_{2y}(x,0)$$
$$E_0^i e^{-jk_1 x \sin\theta_i} + E_0^r e^{-jk_1 x \sin\theta_i} = E_0^t e^{-jk_2 x \sin\theta_t}$$

- Continuity of tangential magnetic field

$$H_{1x}(x,0) = H_{2x}(x,0)$$
$$-\cos\theta_i \frac{E_0^i}{\eta_1} e^{-jk_1 x \sin\theta_i} + \cos\theta_i \frac{E_0^r}{\eta_1} e^{-jk_1 x \sin\theta_i} = -\cos\theta_t \frac{E_0^t}{\eta_2} e^{-jk_2 x \sin\theta_t}$$

Reflection and Transmission Coefficients for Perpendicular Polarization

- Reflection coefficient:

$$\Gamma_\perp = \frac{E_0^r}{E_0^i} = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$$

- Transmission coefficient:

$$\tau_\perp = \frac{E_0^t}{E_0^i} = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$$

Fresnel reflection and transmission coefficients for perpendicular polarization.

Question

- Model a perfect conductor by setting $\eta_2 = 0$ and show that we get back the reflected field from the previous lecture (and that the transmitted field vanishes).

Summary

- A plane wave obliquely incident on the planar boundary between two dielectric half-spaces is partially reflected back into the first medium and partially transmitted into the second medium.
- The reflection and transmission coefficients depend on the polarization of the incident wave quantify how much reflection and transmission occurs.

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